

CN

6 November 2025

IPv4 Addressing and Subnetting Cheat Sheet

For Exam Revision (Beginner-Friendly)

1. What is an IP Address?

An **IP address** uniquely identifies each device on a network. IPv4 uses a **32-bit address** divided into 4 octets.

Binary: 01110101.10010101.00011101.11101010

Dotted Decimal: 117.149.29.234

Each IP has two parts:

IP Address = Network ID (NetID) + Host ID (HostID)

2. IP Address Classes

Class	Starting Bits	Range	Default Mask	CIDR	Usage
A	0	1–126	255.0.0.0	/8	Large Networks
B	10	128–191	255.255.0.0	/16	Medium Networks
C	110	192–223	255.255.255.0	/24	Small Networks

3. Private IP Address Ranges

Class	Private Range	No. of Blocks
A	10.0.0.0 – 10.255.255.255	1
B	172.16.0.0 – 172.31.255.255	16
C	192.168.0.0 – 192.168.255.255	256

4. Subnet Mask Basics

Subnet mask determines which bits represent the network and which represent the hosts.

255.255.255.0 → 11111111.11111111.11111111.00000000

Network Address: All host bits = 0

Broadcast Address: All host bits = 1

Example:

IP = 192.168.1.10, Mask = 255.255.255.0

Network = 192.168.1.0

Broadcast = 192.168.1.255

5. Subnetting a Class C Address

A Class C network has 8 host bits. We can borrow some bits to create subnets.

Binary (4th Octet)	Decimal Mask	CIDR	Subnets	Hosts/Subnet	Block Size
10000000	128	/25	2	126	128
11000000	192	/26	4	62	64
11100000	224	/27	8	30	32
11110000	240	/28	16	14	16
11111000	248	/29	32	6	8
11111100	252	/30	64	2	4

Example: Class C Subnetting

Network: 192.168.10.0, Mask: 255.255.255.128 (/25)

Subnets = $2^1 = 2$, Hosts/Subnet = $2^7 - 2 = 126$

Block Size = $256 - 128 = 128$

Subnet	First Host	Last Host	Broadcast
192.168.10.0	192.168.10.1	192.168.10.126	192.168.10.127
192.168.10.128	192.168.10.129	192.168.10.254	192.168.10.255

6. Subnetting a Class B Address

A Class B network has 16 host bits. You can borrow bits from the 3rd and 4th octets.

CIDR	Subnet Mask	Hosts/Subnet
/17	255.255.128.0	65,534
/18	255.255.192.0	16,382
/19	255.255.224.0	8,190
/20	255.255.240.0	4,094

Example:

Network: 172.16.0.0, Mask: 255.255.128.0

Subnets = $2^1 = 2$, Hosts = $2^{15} - 2 = 65,534$

Block Size = $256 - 128 = 128$

Subnet	First Host	Last Host	Broadcast
172.16.0.0	172.16.0.1	172.16.127.254	172.16.127.255
172.16.128.0	172.16.128.1	172.16.255.254	172.16.255.255

7. Variable Length Subnet Masking (VLSM)

With **VLSM**, different subnet masks can be applied to different subnets to save IP addresses.

Example: Main Network = 192.168.10.0/24

Subnet	Hosts	Block Size	Mask (/CIDR)	Subnet Address	Broadcast
A	70	128	/25	192.168.10.0	192.168.10.127
B	10	16	/28	192.168.10.128	192.168.10.143
C	5	8	/29	192.168.10.144	192.168.10.151
D	30	32	/27	192.168.10.160	192.168.10.191

8. Key Formulas

Number of Subnets = 2^x (x = borrowed bits)

Number of Hosts per Subnet = $2^y - 2$ (y = host bits)

Block Size = $256 - \text{subnet mask value}$

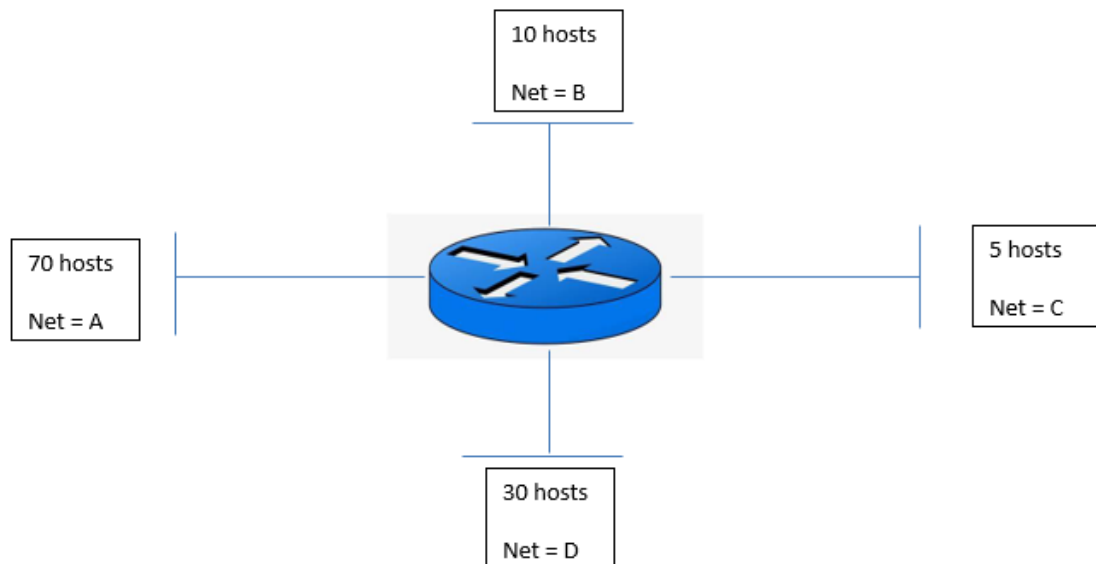
9. Quick Recap Steps

1. Identify the IP class.
2. Write its default subnet mask.
3. Borrow bits as needed.
4. Calculate block size ($256 - \text{mask value}$).
5. List all subnets by adding block size.
6. Identify network, broadcast, and host ranges.

Tip: Always subtract 2 for the network and broadcast addresses when counting valid hosts!

Solved Example (VLSM Practical)

Practice Example #2 for Class C network using VLSM: 192.168.10.0 /24



Create VLSM table for the above four networks. You must have to assign blocks following the name sequence of subnets, that is to assign IP address block to network A first, then B, then C and finally to D.

Subnet	Hosts required	Block size assigned	Subnet Mask (in binary)	Subnet Address	Broadcast Address
A	70	128	11111111 11111111 11111111 10000000	192.168.10.0	192.168.10.127
B	10	16	11111111 11111111 11111111 11110000	192.168.10.128	192.168.10.143
C	5	8	11111111 11111111 11111111 11111000	192.168.10.144	192.168.10.151
D	30	32	11111111 11111111 11111111 11100000	192.168.10.160	192.168.10.191

The block of 32, can start only with multiple of 32 block sizes. That is, it must be either be 0, 32, 64, 96, 128, 160, 192, or 224.

Solved Example (VLSM Practical)

Practice Example for Class C network using VLSM: 192.168.10.0 /24

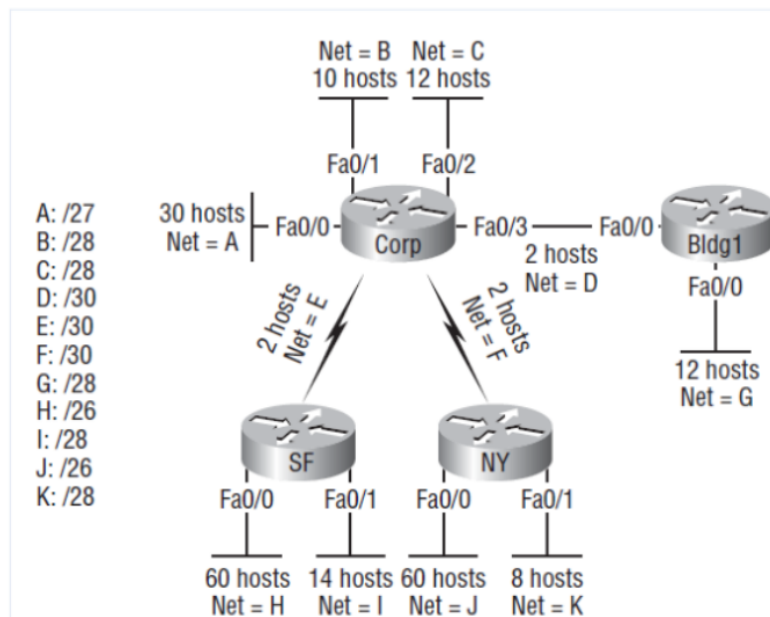


Figure shows a network with 11 networks. Create VLSM table by using block size chart to fill in the table with the subnets you need. This results in two block sizes of 64, one of 32, five of 16, and three of 4.

VLSM Table

Class C Network 192.168.10.0				
Network	Hosts	Block	Subnet	Mask
A	30	32	32	224
B	10	16	0	240
C	12	16	16	240
D	2	4	244	252
E	2	4	248	252
F	2	4	252	252
G	12	16	208	240
H	60	64	64	192
I	14	16	192	240
J	60	64	128	192
K	8	16	224	240

VLSM Sheet (/Chart)

0	
4	
8	
12	B - 192.16.10.0/28
16	
20	
24	C - 192.16.10.16/28
28	
32	
36	
40	
44	
48	A - 192.16.10.32/27
52	
56	
60	
64	
68	
72	
76	
80	
84	
88	
92	
96	H - 192.16.10.64/26
100	
104	
108	
112	
116	
120	
124	
128	
128	
132	
136	
140	
144	
148	
152	
156	
160	J - 192.16.10.128/26
154	
158	
172	
176	
180	
184	
188	
192	
196	
200	I - 192.16.10.192/28
204	
208	
212	
216	G - 192.16.10.208/28
220	
224	
228	
232	K - 192.16.10.224/28
236	
240	
244	D - 192.16.10.244/30
248	E - 192.16.10.248/30
252	F - 192.16.10.252/30
256	

Problem statement (beginner-friendly): We have a single Class C network **192.168.10.0/24** and need to split it into 11 subnets with different host requirements. The required hosts for each subnet are:

Subnet Name	Hosts Needed
A	30
B	10
C	12
D	2
E	2
F	2
G	12
H	60
I	14
J	60
K	8

Step 1: Sort networks by size (largest first)

Why? Because assigning the largest blocks first prevents running out of contiguous space for big subnets — a common beginner mistake.

Subnet	Hosts Needed	Reasonable Block
H	60	64 hosts block (choose /26)
J	60	64 hosts block (choose /26)
A	30	32 hosts block (choose /27)
I	14	16 hosts block (choose /28)
C	12	16 hosts block (choose /28)
G	12	16 hosts block (choose /28)
B	10	16 hosts block (choose /28)
K	8	8 hosts block (choose /29)
D	2	4 hosts block (choose /30)
E	2	4 hosts block (choose /30)
F	2	4 hosts block (choose /30)

Step 2: Understand block sizes and CIDR

A block size is a power of two. For a block with 2^n total addresses, usable hosts = $2^n - 2$ (network + broadcast not usable).

- Block 64 addresses $\Rightarrow$ usable hosts = 62 $\Rightarrow$ CIDR /26.
- Block 32 addresses $\Rightarrow$ usable hosts = 30 $\Rightarrow$ CIDR /27.
- Block 16 addresses $\Rightarrow$ usable hosts = 14 $\Rightarrow$ CIDR /28.
- Block 8 addresses $\Rightarrow$ usable hosts = 6 $\Rightarrow$ CIDR /29.
- Block 4 addresses $\Rightarrow$ usable hosts = 2 $\Rightarrow$ CIDR /30.

Step 3: Allocate subnets left-to-right from 192.168.10.0

Start at 192.168.10.0 and allocate the largest block first.

Subnet	Network	CIDR	First Host	Last Host	Broadcast
H	192.168.10.0	/26	192.168.10.1	192.168.10.62	192.168.10.63
J	192.168.10.64	/26	192.168.10.65	192.168.10.126	192.168.10.127
A	192.168.10.128	/27	192.168.10.129	192.168.10.158	192.168.10.159
I	192.168.10.160	/28	192.168.10.161	192.168.10.174	192.168.10.175
C	192.168.10.176	/28	192.168.10.177	192.168.10.190	192.168.10.191
G	192.168.10.192	/28	192.168.10.193	192.168.10.206	192.168.10.207
B	192.168.10.208	/28	192.168.10.209	192.168.10.222	192.168.10.223
K	192.168.10.224	/29	192.168.10.225	192.168.10.230	192.168.10.231
D	192.168.10.232	/30	192.168.10.233	192.168.10.234	192.168.10.235
E	192.168.10.236	/30	192.168.10.237	192.168.10.238	192.168.10.239
F	192.168.10.240	/30	192.168.10.241	192.168.10.242	192.168.10.243

How these addresses were computed (example for H):

For /26 (block size 64) starting at 192.168.10.0:

- Network = 192.168.10.0 (host bits all zero).
- Broadcast = network + 63 = 192.168.10.63.
- First usable host = network + 1 = 192.168.10.1.
- Last usable host = broadcast - 1 = 192.168.10.62.

So the subnet size is 64 addresses.

Binary Representation

Network: 192.168.10.0/26

We only analyze the last octet:

$$192.168.10.\mathbf{0} = 00000000 \text{ (binary)}$$

6 host bits means host range varies:

- Lowest host value: 000000 (binary)
- Highest host value: 111111 (binary)

Convert 111111 to decimal:

$$111111 = 1 * 2^5 + 1 * 2^4 + 1 * 2^3 + 1 * 2^2 + 1 * 2^1 + 1 * 2^0 = 32 + 16 + 8 + 4 + 2 + 1 = 63$$

This is why we add 63 for the broadcast address.

Final Address Computation

- Network Address = 192.168.10.0
- Broadcast Address = 192.168.10.0 + 63 = 192.168.10.63
- First Usable Host = 192.168.10.1
- Last Usable Host = 192.168.10.62

Step 4: Verify no overlaps

Each block starts at the previous block's broadcast + 1, so there are no overlaps. After the last allocated /30, the next free address would be 192.168.10.244 (if you want to check remaining space).

Beginner Tips and Common Pitfalls

- **Allocate largest subnets first.**
- **Always subtract 2** (network + broadcast) when counting usable hosts.
- **Block arithmetic:** For last-octet mask value M, block size = 256 - M. Example: /26 mask is 255.255.255.192 so block size = 64.
- **When unsure, convert to binary** to see boundaries.
- **Label network, first host, last host, broadcast** in answers for exams.

Final note: VLSM is powerful — sort requirements, choose block sizes that fit, allocate from the start of the range, and always verify no overlaps.